CS 210

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Program 2 Analysis

1. Briefly define a *queue* data structure.

   A queue is a FIFO structure. The first request added is the first request removed.

   What evidence did you find that a *queue* container was operating in the load balancer?

   The queue removes requests with **front()** and then **pop()**. The output begins with older queued requests, including PID 1 before later requests.

2. Briefly define a *stack* data structure.

   A stack is a LIFO structure. The last request added is the first request removed.

   What evidence did you find that a *stack* container was operating in the load balancer?

   The stack uses **top()** and **pop()**. Its first completed PIDs were 7, 9, 17, 10, and 8, showing newer requests were selected first.

3. Briefly define a *priority queue* data structure.

   A priority queue removes the request with the highest priority according to its comparator instead of simply using arrival order.

   What evidence did you find that a *Max priority queue based on process time* was operating in the load balancer?

   The first process times completed were 362, 395, 399, 426, and 480. These large process times show that the MaxProcessTime comparator selected long requests first.

   What evidence did you find that a *Min priority queue based on process time* was operating in the load balancer?

   The output started with short process times such as 84, 87, and 31. The MinProcessTime version completed 399 requests, showing that short requests were selected first.

   What evidence did you find that a *Max priority queue based on process ID* was operating in the load balancer?

   The first completed PIDs were 7, 9, 17, 10, and 8 instead of the older PIDs. This shows that the MaxPID comparator selected the newest or highest-PID requests first. Its output matched the stack because both selected the newest request.

Evidence can be a description of the output specific to the data structure and/or an annotated image.